Fiscal Year:	FY 2022	**Task Last Updated:**	FY 11/10/2022
PI Name:	Wegst, Ulrike Ph.D.		
Project Title:	Structure-Property-Processing Correlations in Freeze-Cast Biomimetic Materials: 80NSSC21K0039		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Comments:	NOTE: PI moved to Northeastern University from Dartmouth College in summer 2020.		
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
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No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Erb, Randall Ph.D. (Northeastern University) Karma, Alain Ph.D. (Northeastern University)		
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Task Description:	NOTE: Continuation of "Structure-Property-Processing Correlations in Freeze-Cast Biomimetic Materials" (grant 80NSSC18K0305 at Dartmouth College), with the same Principal Investigator, Dr. Ulrike Wegst, due to PI move to Northeastern University in summer 2020. Strong, tough, and lightweight materials are needed for a myriad of structural and functional applications in Space systems and on Earth in diverse strategic fields that range from biomedical, to buildings, transportation and energy generation, or storage. These new, yet to be developed materials, should exhibit unprecedented combinations of strength and toughness, while being manufacturable at low cost and in high volume. Numerous processing techniques have been proposed to produce these materials, but frequently cannot provide the degree of microstructural control needed to manipulate and optimize the property profile, particularly of multifunctional materials. One technique, which holds great promise in fabricating novel materials with great structural control, is freeze-casting, a relatively inexpensive procedure which also provides a means to mimic complex, efficient natural materials with hierarchical designs over several length-scales. To date, the fundamental science and understanding of this processing technique is limited, because, on Earth, convection and sedimentation complicate both the interpretation of experimental results and their comparison with theoretical predications. Avoiding these in space, following four scientific goals will be pursued. To reveal on Earth (Aim 1) and in Space (Aim 2) the underlying fundamental materials science of freeze casting and establish a realistic model for structure formation in multifaceted crystal nucleation and growth. Aim 3: Develop predictive models to identify the most promising material processing combinations and conditions to optimize material performance. Aim 4: Translate the results of (1-3) into structure-property-processing correlations for highly porous, freeze-cast scaffolds (polymer, ceramic, metal, and composite) made on Earth. Data obtained and computational models will be made available to the NASA Physical Sciences Informatics to make the data accessible to the global community and help translate research into application for an accelerated pace of advanced-materials discovery, innovation, manufacture, and commercialization.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our goal is to apply the fundamental knowledge gained through this study in the development of new and improved freeze-cast (ice-templated) materials that are strong, tough and lightweight and can be custom-designed for a given application. Such materials are required for applications that range from biomedical and tissue regeneration to filtration, catalysis, transportation, and energy generation and storage. Only with fundamental, physics-based models, systematic materials design and synthesis will this be possible. The results of the terrestrial and microgravity research will be collated, documented, and captured also in a visual format such as in, for example, structure-property-processing charts. With these we wish to enable a paradigm shift towards a more systematic approach in materials synthesis by pointing into particularly promising materials research directions.
Task Progress:	When aqueous solutions or slurries without additives are freeze-cast, meaning directionally solidified, the ice crystals grow with their basal plane oriented parallel to the temperature gradient and along the crystallographic <11-20> direction. Because of this orientation, the structure of the typical wine-cork-like freeze-cast sample is lamellar once quasi-steady state solidification has been reached. The first two layers in a sample are dense and cellular in structure, only a few 100 µm thick, and initially formed during the first seconds of solidification, with cooling rates so high that no phase separation occurs, but the dissolved/suspended material is trapped within the ice crystals. In some cases, when water-based biopolymer solutions are freeze-cast, the final freeze-cast materials and scaffolds exhibit features that are very unusual for directionally solidified materials; they exhibit, for example: i) highly aligned and regularly spaced ridges of uniform height, ‘jellyfish’-like caps and arrays, and ‘tentacles’-like features parallel to the direction of solidification, and ii) fibrillar pillars and bridges in fiber composites, and even enhanced iii) fibrillation, a form of self-assembly and structure formation in biopolymers such as collagen. In our study of “Structure-Property-Processing Correlations in Freeze-Cast Biomimetic Materials,” we quantify both the (to date, little studied) complex, dynamic ice crystal growth phenomena and the ice-templated hierarchical architecture of freeze-cast scaffolds, as well as the various cell wall surface features, not only because they are of interest from a fundamental science point of view but also because the ridges, pillars, bridges, and fibrillation observed in freeze-cast biopolymer scaffolds frequently result in highly desirable features for biomedical applications. In peripheral nerve repair, we observe, for example, that not only the highly aligned porosity, but also cell wall surface ridges serve as powerful structural cues to guide regenerating neurites from one end to the other of the freeze-cast regenerative scaffold. While cell wall surface features, such as these regularly spaced ridges, had been observed under specific processing conditions in a range of freeze-cast biopolymers, we found that the fundamental science and underlying principles that govern their formation remain unknown and that, in fact, more generally, ice crystal growth from the liquid phase and binary or ternary polymer solutions had hardly ever been studied and simulated. We therefore started to systematically study structure-property-processing correlations under well-defined conditions and complement experiments with phase-field (PF) simulations, which are essential to fully understand the observed phenomena and formulate robust and predictive structure-property-processing correlations for the design and manufacture of freeze-cast materials for a great variety of applications that range from biomedical to energy generation and storage, to catalysis, filtration, optical and acoustical absorption, and thermal insulation. In the first instance, we experimentally determined structure-property-processing correlations for scaffolds freeze-cast from collagen- and chitosan-based scaffolds, both in the dry and the fully hydrated states. The results obtained for both ‘smooth’ and ‘fibrillar’ polymer systems contributed to a better understanding of the likely sequence of events during the solidification process and the resulting ice-templated feature formation in the polymer phase. Noteworthy are also discoveries made with respect to the dependence of the mechanical properties, such as modulus, yield strength, and toughness, on processing conditions such as applied cooling rates and mold height, and the resulting structural features such as lamellar spacing (length of the short pore axis of the rectangular pores), and cell wall thickness, and scaffold reinforcing features such as ridges, bridges, and pillars, and preferential molecular alignment with the freezing direction. These systematic experimental studies were key first steps on the path to our long-term goal of this study: to bridge the current critical knowledge gap, which is the lack of understanding of the fundamental mechanisms of structure formation, initially in freeze-cast polymers, and later also in particle-based ceramic and metal systems. Knowing how the size and shape of the pores in freeze-cast materials are defined and can be controlled by controlling the ice crystal growth, the need for a better fundamental understanding of the physics of directional solidification in aqueous systems becomes even more obvious. We overcame the challenge of the absence of equilibrium phase diagrams and, even more important, the non-equilibrium state diagrams, which describe the metastable ‘states’ fundamental to an understanding

of these systems; we determined concentration- and temperature-dependent diffusion coefficients and viscosities for our ternary chitosan-acetic acid-water system; and further reduced system complexity in support of the PF simulation effort by focusing our 2D and 3D studies on the binary systems trehalose-water and sucrose-water systems, for which the fundamental properties are known and documented.

Since crystal growth in bulk samples are difficult to observe and characterize in situ, diffusive crystal growth under terrestrial conditions is typically studied in 2D with thin samples. These thin samples are designed so that the solidification conditions can be controlled carefully and only a single, two-dimensional (2D) array of dendrites (or cells) forms, and the solid-liquid interface, interface dynamics, and the formation of instabilities can be observed. Initially (in Years 1-3), we focused on the traditional light microscopy perpendicular to the moving freezing front primarily in the trehalose-water and sucrose-water systems. To observe the formation of Mullins-Sekerka instabilities and their subsequent reorganization into an array of cells or dendrites as a function of polymer concentration and cooling rates (which is equal to the product of velocity and thermal gradient in directional solidification), we discovered X-ray tomoscopy, which is continuous, time-resolved tomography, as the more suitable tool for crystal growth phenomena.

With synchrotron-based tomoscopy, we have in Years 3 and 4 been able to investigate and determine the kinetics of the ice-phase growth and the mechanisms by which the growing ice crystals (thus ice phase), templates the polymer phase into a cellular solid with a range of cell wall surface features, while the polymer is increasingly upconcentrated and transitions from an aqueous solution to a glassy solid. Overall, our observation is that 2D thin film experiments should, whenever possible, be complemented by 3D in situ tomoscopy experiments, because it is impossible to extrapolate structure formation phenomena from thin film observations to bulk materials, in which the third dimension and grain or domain interactions play an important role.

As critical as 3D in situ X-ray tomoscopy has become for our study for the quantitative 3D imaging and observation also of transient events during the directional solidification in freeze-casting, are the 3D PF simulations, which started in Year 2. Remarkably, the PF simulations of anisotropic, faceted ice crystal growth binary aqueous systems with a dilute impurity (trehalose, sucrose), reproduced for the first time the various salient features of freeze-cast structures: from the cellular scaffold structure with their characteristic lamellar spacing to the cell wall surface features such as ridges, which are templated by secondary instabilities on the ice crystal's basal plane. The value of the 3D phase-field simulations is immense, since they enable us to investigate little studied and not yet modeled mechanisms of hierarchical microstructure formation and cell wall material self-assembly, both of which define anisotropy, property profile, and overall performance of freeze-cast materials. Our results highlight the crucial role of both the solid-liquid interface free-energy and the interface kinetics in structure formation. They also highlight that other potentially important physical effects neglected in the phase-field model, such as fluid flow generated by the expansion of ice, may play a more secondary role in structure formation.

From a practical point of view, the ability to accurately predict the detailed temporal evolution of interface structures under prescribed growth conditions is critical for optimizing processing routes and material properties.

Bibliography Type:	Description: (Last Updated: 07/29/2025)
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